

Classical-Type Dynamical Lie Algebras of Open $2 \times L$ Ising Layers

Exact 3D Ising Research Program
Draft accompanying the wave-17 artifact bundle

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Abstract

We determine the characteristic-zero dynamical Lie algebras generated by the local Pauli terms of open $2 \times L$ Ising layers at $L = 2, 3, 4, 5$, and prove a uniform all- L containment theorem. The local-term algebra is distinct from the smaller two-generator algebra generated only by $A = \sum X_i$ and $B = \sum ZZ$. Exact Pauli closure, parity splitting, invariant signed checkerboard forms, and rational rank certificates give

$$\mathfrak{g}_{2 \times 2} = \mathfrak{so}_8(\mathbb{Q}) \oplus \mathfrak{so}_8(\mathbb{Q}), \quad \mathfrak{g}_{2 \times 3} = \mathfrak{sp}_{32}(\mathbb{Q}) \oplus \mathfrak{sp}_{32}(\mathbb{Q}),$$

$$\mathfrak{g}_{2 \times 4} = \mathfrak{so}_{128}(\mathbb{Q}) \oplus \mathfrak{so}_{128}(\mathbb{Q}), \quad \mathfrak{g}_{2 \times 5} = \mathfrak{sp}_{512}(\mathbb{Q}) \oplus \mathfrak{sp}_{512}(\mathbb{Q}).$$

For every $L \geq 2$, the generators lie in the corresponding orthogonal or symplectic product and satisfy an explicit dimension upper bound. Equality, saturation, maximality, and type classification are asserted only at $L \leq 5$; the all- L reverse-inclusion problem remains open.

1 The object and a necessary distinction

Let $n = 2L$ and let $\mathfrak{g}_L$ be the dynamical Lie algebra over $\mathbb{Q}$ generated by all local terms of the open ladder:

$$\{X_i : 1 \leq i \leq n\} \cup \{Z_u Z_v : (uv) \text{ is an open ladder bond}\}.$$

There are n single-site terms and $3L - 2$ bonds. The algebra acts on $V = \mathbb{Q}^{2^n}$.

This paper studies the local-term algebra. It must not be confused with the two-generator algebra $\langle A, B \rangle_{\text{Lie}}$, where $A = \sum_i X_i$ and $B = \sum_{(uv)} Z_u Z_v$. For the open 2×3 layer the former has dimension 1056 and type $\mathfrak{sp}_{32} \oplus \mathfrak{sp}_{32}$, whereas the latter has dimension 263 and a different semisimple quotient. Modular $\mathfrak{sp}_{14}$, $\mathfrak{sp}_6$, and $\mathfrak{gl}_6$ witnesses belong to that 263-dimensional two-generator algebra and are not quotients of the 1056-dimensional local-term algebra.

All Pauli brackets are exact. In the binary encoding $(x \mid z) \in \mathbb{F}_2^{2n}$, a nonzero bracket is the XOR product, up to a nonzero scalar. Therefore exact BFS closure by Pauli strings gives a rational basis and an exact dimension, with no floating-point rank inference.

2 Exact characteristic-zero classifications

2.1 $L = 2$: orthogonal base case

The open 2×2 layer has $n = 4$, two product- X parity halves of dimension 8, and

$$\dim \mathfrak{g}_2 = 56 = 2 \dim \mathfrak{so}_8.$$

The exact invariant form on each half is symmetric; the local-term algebra is

$$\mathfrak{g}_2 \cong \mathfrak{so}_8(\mathbb{Q}) \oplus \mathfrak{so}_8(\mathbb{Q}).$$

This is the finite control for the alternation below.

2.2 $L = 3$: first symplectic point

Theorem 2.1 (Exact 2×3 local-term algebra). *For the six single-site terms and seven open bonds of the 2×3 layer,*

$$\dim_{\mathbb{Q}} \mathfrak{g}_3 = 1056, \quad \mathfrak{g}_3 \cong \mathfrak{sp}_{32}(\mathbb{Q}) \oplus \mathfrak{sp}_{32}(\mathbb{Q}).$$

The algebra is perfect, its diagonal Killing form is nondegenerate with exact signature $(544, 512)$, and it splits into two 528-dimensional ideals selected by the two product- X parity representations. Each ideal acts faithfully on one 32-dimensional parity half-module, which carries a nondegenerate alternating rational form.

The exact Pauli closure reaches 1056 strings. The two product- X sectors of the 64-dimensional Hilbert space each have dimension 32; the two corresponding ideals each have dimension 528. Exact nonzero minors at two primes certify the two simple ideals. The chain and 2×2 controls recover $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$, $\mathfrak{sl}_4$, and $\mathfrak{so}_8 \oplus \mathfrak{so}_8$ in the expected small cases.

2.3 $L = 4$: orthogonal alternation

Theorem 2.2 (Exact 2×4 local-term algebra). *For the eight single-site terms and ten open bonds of the 2×4 layer,*

$$\dim_{\mathbb{Q}} \mathfrak{g}_4 = 16256, \quad \mathfrak{g}_4 \cong \mathfrak{so}_{128}(\mathbb{Q}) \oplus \mathfrak{so}_{128}(\mathbb{Q}).$$

The algebra is perfect, has nondegenerate diagonal Killing signature $(8192, 8064)$, and splits on the two product- X parity halves. Each half preserves a nondegenerate symmetric rational form of determinant one, and its image is the full corresponding orthogonal algebra.

The tempting $\mathfrak{sp}_{128} \oplus \mathfrak{sp}_{128}$ analogy is false: the invariant form is symmetric at even L . The two-generator 2×4 algebra is a separate finite object of dimension 2952 and is not the algebra classified here.

2.4 $L = 5$: exact characteristic-zero symplectic point

Theorem 2.3 (Exact 2×5 local-term algebra). *For the 23 local terms of the open 2×5 layer,*

$$\mathfrak{g}_5 \cong \mathfrak{sp}_{512}(\mathbb{Q}) \oplus \mathfrak{sp}_{512}(\mathbb{Q}), \quad \dim \mathfrak{g}_5 = 262656.$$

The two ideals act on the product- X parity halves of dimension 512. Each half carries a nondegenerate alternating signed-checkerboard invariant form, unique up to scalar.

The proof reaches exactly 262656 Pauli strings by integer BFS. The set-law count is

$$(2^9 - 1)2^9 + 2^{10} = 262656,$$

and every reached string satisfies the law. The full 134,479,872-equation sign system has zero contradictions and one connected component, giving existence and uniqueness of the alternating form up to scalar. This is an exact characteristic-zero theorem, not merely the earlier modular probe.

L	n	$\dim \mathfrak{g}_L$	exact type
2	4	56	$\mathfrak{so}_8 \oplus \mathfrak{so}_8$
3	6	1056	$\mathfrak{sp}_{32} \oplus \mathfrak{sp}_{32}$
4	8	16256	$\mathfrak{so}_{128} \oplus \mathfrak{so}_{128}$
5	10	262656	$\mathfrak{sp}_{512} \oplus \mathfrak{sp}_{512}$

Table 1: Certified local-term algebras over $\mathbb{Q}$.

3 Uniform all- L containment theorem

Let $J = (1, \dots, 1) \in \mathbb{F}_2^n$ and let d be one checkerboard class of the ladder. Define

$$S_L = \{(x \mid z) : z \text{ has even weight, } z \neq J, (J - z) \cdot x = 1 + z \cdot d\}$$

with, when L is odd, the additional full fibre $\{(x \mid J) : x \in \mathbb{F}_2^n\}$. On each product- X parity half, pair the class labels by the signed checkerboard matching $s \leftrightarrow s + d$.

Lemma 3.1 (Uniform forward closure). *For every $L \geq 2$, the set S_L is closed under every nonzero adjoint action of an X_i or a ladder bond $Z_u Z_v$. Hence*

$$\mathfrak{g}_L \subseteq \text{span}_{\mathbb{Q}}(S_L).$$

The proof has three cases. An X_i action preserves the affine fibre; a bond action preserves the law because each ladder bond crosses the checkerboard; and the exceptional $z + z_g = J$ case either has zero commutator for even L or lands in the full $z = J$ fibre for odd L .

Theorem 3.2 (All- L invariant-form containment). *For every $L \geq 2$, $n = 2L$,*

$$\mathfrak{g}_L \subseteq \begin{cases} \mathfrak{so}_{2^{n-1}}(\mathbb{Q}) \oplus \mathfrak{so}_{2^{n-1}}(\mathbb{Q}), & L \text{ even,} \\ \mathfrak{sp}_{2^{n-1}}(\mathbb{Q}) \oplus \mathfrak{sp}_{2^{n-1}}(\mathbb{Q}), & L \text{ odd,} \end{cases}$$

with respect to the signed checkerboard form on the two product- X halves. Consequently,

$$\dim \mathfrak{g}_L \leq 2^{n-1}(2^{n-1} - (-1)^L).$$

At $L = 2, 3, 4, 5$ equality holds, and the exact classifications in Section 2 follow. Saturation/equality for all $L > 5$ is not proved.

The form-membership condition is the same parity equation as the set law:

$$|x| = 1 + z \cdot x + z \cdot d \pmod{2}.$$

The signed matching obeys $\Omega_L^T = (-1)^L \Omega_L$, so it is symmetric for even L and alternating for odd L . At the four certified points, dimension equality forces the full classical algebra on each half and the sign-system calculation gives uniqueness of the form.

4 What remains open

The all- L reverse inclusion $\text{span}(S_L) \subseteq \mathfrak{g}_L$ is the central missing step. The reviewed path-telescoping construction generates only a four-parameter family per pair, not arbitrary fibre patterns. Thus $L =$

6 and beyond are predictions, not classifications: for example, $L = 6$ would predict $\mathfrak{so}_{2048} \oplus \mathfrak{so}_{2048}$ of dimension 4192256, but no exact-Q saturation theorem is claimed here.

Other unresolved issues include a uniform proof of form uniqueness, rational isotropy questions outside the explicit split forms, and the relation of this local-term family to the distinct two-generator transfer algebra. The 3×3 layer is outside the $2 \times L$ alternation family and enters a different regime.

These finite Lie-algebra classifications do not diagonalize the 3D Ising transfer matrix or solve its thermodynamic limit. They are structural data and exact constraints on any proposed integrable mechanism.

Artifact and reproduction map

- 2×3 : `results/algebra/char0_quotient.json`, `experiments/e143_char0_quotient.py`, `tests/test_char0_quotient.py`; the historical source note `proofs/char0_quotient.md` is referenced by the report but is not present in the current proof directory.
- 2×4 : `proofs/char0_2x4.md`, `results/algebra/char0_2x4.json`, `experiments/e148_char0_2x4.py`, `tests/test_char0_2x4.py`.
- 2×5 : `proofs/alternation_l5_char0.md`, `results/algebra/alternation_l5_char0.json`.
- All- L containment: `proofs/alternation_law.md`, `results/algebra/alternation_law.json`, `tests/test_alternation_law.py`.
- General Pauli algebra conventions: `src/ising/clifford.py`, `results/dla_table.json`.

All dimensions and form statements can be regenerated from the repository interpreter and standalone tests. The acceptance suite recorded in the bundle is a separate integration gate and does not replace these targeted checks.